



Republic of the Philippines
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College of Informatics and Computing Sciences

Pointers to Review

System Thinking Midterm Exam Reviewer

I. Core Concepts: Definition, History, and Characteristics of a System

A. The Definition of a System

- Understand the fundamental definition: A system is a set of interacting or interdependent components forming an integrated whole.
- A system involves elements (components) and relationships that are different from the relationships to external elements or sets.
- Fields that study general properties of systems include Systems science, systems theory, cybernetics, dynamical systems, thermodynamics, and complex systems.

B. Common Characteristics of a System

- Structure: It contains parts (or components) that are directly or indirectly related to each other.
- Behavior: It contains processes that transform inputs into outputs (material, energy, or data).
- Interconnectivity: The parts and processes are connected by structural and/or behavioral relationships.
- Systems can be broken down into subsystems and sub-processes.

C. Historical Context and Etymology

- Know the origin of the term "system" from the Latin *systema*, and the Greek συστημα (*systema*), meaning "whole compounded of several parts or members, system," or "composition".
- Know the earlier meaning: "total," "crowd," or "union," derived from the verb *sunistemi* (uniting, putting together).
- Identify

Nicolas Léonard Sadi Carnot was the first to develop the concept of a "system" in the natural sciences, studying the "working substance" in thermodynamics (steam engines).

- Identify

Rudolf Clausius for generalizing this picture to include the concept of the "surroundings" and using the term "working body" for the system.

II. Systems Thinking Philosophy

- Definition: Systems thinking is the process of understanding how things, regarded as systems, influence one another within a whole.

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- Core Goal: It is an approach to problem solving by viewing "problems" as parts of an overall system, rather than reacting to specific parts, outcomes, or events.
- Holistic View: It proposes to view systems in a holistic manner, in contrast to Descartes's scientific reductionism.
- Relationships over Isolation: It is based on the belief that component parts are best understood in the context of their relationships with each other and with other systems, rather than in isolation.
- Causality: It focuses on cyclical rather than linear cause and effect.
- Interdependence: Know that systems thinking acknowledges that an improvement in one area can adversely affect another area. This promotes organizational communication to avoid the "silo effect".

III. Causal Loop Diagrams (CLD) and Feedback

A. Causal Links and Variables

- A CLD is a causal diagram that visualizes how interrelated variables (nodes) affect one another.
- The relationship between variables (arrows) can be labeled as positive (+) or negative (-).
 - Positive Causal Link (+): The two nodes change in the same direction. If the starting node increases, the other node increases; if the starting node decreases, the other node decreases.
 - Negative Causal Link (-): The two nodes change in opposite directions. If the starting node increases, the other node decreases, and vice versa.
- Delay: Be aware of how delays are represented (a short line across the causal link) and their effect (the system might fluctuate).

B. Feedback Loops (Reinforcing vs. Balancing)

- A chain of cause and effect exists where a variable has a feedback loop influencing its own rate of change.
- Determining Loop Type (The Test): Start with an assumption ("Node 1 increases") and follow the loop. The loop is:
 - Reinforcing (R): If, after going around the loop, the result is the same as the initial assumption (an increase leads to a further increase).
 - Balancing (B): If the result contradicts the initial assumption (an increase leads to a decrease, or vice versa).
- Mathematical Rule of Signs: Count the number of negative (-) links in the loop:
 - Even number of - signs (or zero): Reinforcing Loop (R).
 - Odd number of - signs: Balancing Loop (B).

C. Loop Behavior



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- Reinforcing Loops (R/Positive Feedback):
 - Associated with exponential increases/decreases (growth or collapse).
 - The effects of a small disturbance include an increase in the magnitude of the perturbation.
 - Leads to amplification of the original signal.
 - Any system with positive feedback and a gain greater than one will result in a runaway situation.
- Balancing Loops (B/Negative Feedback):
 - Associated with reaching a plateau or a goal/target.
 - The result of a process influences the operation to reduce changes.
 - Tends to make a self-regulating system and produces stability.

IV. Application and Diagramming Skills

- Task: Successfully analyze the relationship between 4 variables given in a specific sequence to form either an R or B loop.
- Review Strategy: Practice identifying the +, – signs for the four-causal links in the required sequence. Then, apply the Rule of Signs (or the "start with an assumption" test) to confirm if the resulting loop is R or B.
- Task: Design your own complete Causal Loop Diagrams (CLDs) consisting of at least one R loop and one B loop, based on a case topic and a specific viewpoint.
- Review Strategy:
 1. Define Nodes: Select relevant variables (nodes) for the specific corruption problem (e.g., *Level of Audit Scrutiny*, *Perceived Risk of Getting Caught*, *Frequency of Corrupt Acts*, *Public Outcry*). Ensure at least four are used in each loop.
 2. Philippine Corruption Topics: Activity
 3. Labeling: Double-check that all links (+/–) and loop types (R/B) are correctly placed and logically sound according to the Rule of Signs.